Year 10 Higher

REVISION PACK

Progress Check 1

ANSWERS

Units in this pack

- Number 1
- Algebra 1

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 1 · Working with fractions (Lesson 1.1)

Q	Working	Answer
adding fractions		
1.1	$\frac{12}{15} + \frac{10}{15} = \frac{22}{15}$ $= 1\frac{7}{15}$	$1\frac{7}{15}$
1.2	$\frac{20}{24} + \frac{9}{24} = \frac{29}{24}$ $= 1\frac{5}{24}$	$1\frac{5}{24}$
1.3	$\frac{9}{12} + \frac{8}{12} = \frac{17}{12}$ $= 1\frac{5}{12}$	$1\frac{5}{12}$
1.4	$\frac{10}{14} + \frac{7}{14} = \frac{17}{14}$ $= 1\frac{3}{14}$	$1\frac{3}{14}$
1.5	$\frac{21}{24} + \frac{8}{24} = \frac{29}{24}$ $= 1\frac{5}{24}$	$1\frac{5}{24}$
1.6	$\frac{10}{18} + \frac{15}{18} = \frac{25}{18}$ $= 1\frac{7}{18}$	$1\frac{7}{18}$
multiplying mixed numbers		
1.7	$\frac{3}{2} \times \frac{8}{3}$ $= \frac{24}{6}$ $= 4$	4

1.8	$\frac{10}{3} \times \frac{8}{5}$ $= \frac{80}{15}$ $= \frac{16}{3}$ $= 5\frac{1}{3}$	$5\frac{1}{3}$
1.9	$\frac{9}{4} \times \frac{11}{6}$ $= \frac{99}{24}$ $= \frac{33}{8}$ $= 4\frac{1}{8}$	$4\frac{1}{8}$
1.10	$\frac{7}{4} \times \frac{12}{5}$ $= \frac{84}{20}$ $= \frac{21}{5}$ $= 4\frac{1}{5}$	$4\frac{1}{5}$
1.11	$\frac{8}{3} \times \frac{9}{8}$ $= \frac{72}{24}$ $= 3$	3
1.12	$\frac{15}{4} \times \frac{5}{3}$ $= \frac{75}{12}$ $= \frac{25}{4}$ $= 6\frac{1}{4}$	$6\frac{1}{4}$
dividing mixed numbers		

1.13	$\frac{9}{2} \div \frac{6}{5}$ $= \frac{9}{2} \times \frac{5}{6}$ $= \frac{45}{12}$ $= \frac{15}{4}$ $= 3\frac{3}{4}$	$3\frac{3}{4}$
1.14	$\frac{8}{3} \div \frac{4}{3}$ $= \frac{8}{3} \times \frac{3}{4}$ $= \frac{24}{12}$ $= 2$	2
1.15	$\frac{15}{4} \div \frac{5}{2}$ $= \frac{15}{4} \times \frac{2}{5}$ $= \frac{30}{20}$ $= \frac{3}{2}$ $= 1\frac{1}{2}$	$1\frac{1}{2}$
1.16	$\frac{21}{4} \div \frac{7}{4}$ $= \frac{21}{4} \times \frac{4}{7}$ $= \frac{84}{28}$ $= 3$	3
1.17	$\frac{12}{5} \div \frac{6}{5}$ $= \frac{12}{5} \times \frac{5}{6}$ $= \frac{60}{30}$ $= 2$	2

1.18	$\frac{25}{6} \div \frac{5}{3}$ $= \frac{25}{6} \times \frac{3}{5}$ $= \frac{75}{30}$ $= \frac{5}{2}$ $= 2\frac{1}{2}$	$2\frac{1}{2}$
subtracting with a mixed number		
1.19	$\frac{7}{3} - \frac{5}{6}$ $\frac{14}{6} - \frac{5}{6} = \frac{9}{6}$ $= \frac{3}{2}$ $= 1\frac{1}{2}$	$1\frac{1}{2}$
1.20	$\frac{8}{5} - \frac{3}{4}$ $\frac{32}{20} - \frac{15}{20} = \frac{17}{20}$	$\frac{17}{20}$
1.21	$\frac{13}{4} - \frac{2}{3}$ $\frac{39}{12} - \frac{8}{12} = \frac{31}{12}$ $= 2\frac{7}{12}$	$2\frac{7}{12}$
1.22	$\frac{12}{5} - \frac{5}{6}$ $\frac{72}{30} - \frac{25}{30} = \frac{47}{30}$ $= 1\frac{17}{30}$	$1\frac{17}{30}$
1.23	$\frac{11}{6} - \frac{3}{4}$ $\frac{22}{12} - \frac{9}{12} = \frac{13}{12}$ $= 1\frac{1}{12}$	$1\frac{1}{12}$

1.24	$\frac{5}{2} - \frac{4}{5}$ $\frac{25}{10} - \frac{8}{10} = \frac{17}{10}$ $= 1\frac{7}{10}$	$1\frac{7}{10}$
fraction problems		
1.25	$1 - \frac{2}{5} = \frac{3}{5}$ $\frac{1}{3} \times \frac{3}{5} = \frac{1}{5}$ $\frac{3}{5} - \frac{1}{5} = \frac{2}{5}$	$\frac{2}{5}$
1.26	$1 - \frac{1}{4} = \frac{3}{4}$ $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$ $\frac{3}{4} - \frac{1}{2} = \frac{1}{4}$	$\frac{1}{4}$
1.27	$1 - \frac{3}{8} = \frac{5}{8}$ $\frac{1}{5} \times \frac{5}{8} = \frac{1}{8}$ $\frac{5}{8} - \frac{1}{8} = \frac{1}{2}$	$\frac{1}{2}$
1.28	$1 - \frac{1}{6} = \frac{5}{6}$ $\frac{3}{5} \times \frac{5}{6} = \frac{1}{2}$ $\frac{5}{6} - \frac{1}{2} = \frac{1}{3}$	$\frac{1}{3}$
1.29	$1 - \frac{3}{7} = \frac{4}{7}$ $\frac{1}{2} \times \frac{4}{7} = \frac{2}{7}$ $\frac{4}{7} - \frac{2}{7} = \frac{2}{7}$	$\frac{2}{7}$
1.30	$1 - \frac{2}{9} = \frac{7}{9}$ $\frac{3}{7} \times \frac{7}{9} = \frac{1}{3}$ $\frac{7}{9} - \frac{1}{3} = \frac{4}{9}$	$\frac{4}{9}$

Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)		
Q	**Working**	**Answer**
order of operations		
2.1	**(a)** $5^2 = 25$ $2 \times 25 = 50$ $7 + 50 = 57$ **(b)** $(3 + 5) \times 4 - 2 = 30$	**(a) 57** **(b)** $(3 + 5) \times 4 - 2 = 30$
2.2	**(a)** $2^3 = 8$ $3 \times 8 = 24$ $20 - 24 = -4$ **(b)** $2 \times (6 + 3) - 1 = 17$	**(a) −4** **(b)** $2 \times (6 + 3) - 1 = 17$
2.3	**(a)** $3^2 = 9$ $18 \div 9 = 2$ $6 + 2 = 8$ **(b)** $4 + 8 \times (2 + 1) = 28$	**(a) 8** **(b)** $4 + 8 \times (2 + 1) = 28$
2.4	**(a)** $2^3 = 8$ $4 \times 8 = 32$ $9 + 32 = 41$ **(b)** $5 \times (4 - 1) + 2 = 17$	**(a) 41** **(b)** $5 \times (4 - 1) + 2 = 17$
2.5	**(a)** $2^2 = 4$ $6 \times 4 = 24$ $3 + 24 = 27$ **(b)** $(10 - 2) \times 3 + 1 = 25$	**(a) 27** **(b)** $(10 - 2) \times 3 + 1 = 25$
2.6	**(a)** $3^2 = 9$ $5 \times 9 = 45$ $40 - 45 = -5$ **(b)** $(6 + 2) \times 7 - 3 = 53$	**(a) −5** **(b)** $(6 + 2) \times 7 - 3 = 53$
rounding		
2.7		**(a) 275.8** **(b) 280**

2.8		**(a) 1093.5** **(b) 1100**
2.9		**(a) 48.1** **(b) 48**
2.10		**(a) 736.3** **(b) 740**
2.11		**(a) 5281.6** **(b) 5300**
2.12		**(a) 20.0** **(b) 20**

Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

Q	Working	Answer
substitution into v = u + at		
3.1	$v = 3 + (-4) \times 2$ $= 3 - 8$ $= -5$	$v = -5$
3.2	$v = 10 + (-3) \times 6$ $= 10 - 18$ $= -8$	$v = -8$
3.3	$v = -4 + 5 \times 3$ $= -4 + 15$ $= 11$	$v = 11$
3.4	$v = 20 + (-6) \times 3$ $= 20 - 18$ $= 2$	$v = 2$
3.5	$v = -2 + (-3) \times 4$ $= -2 - 12$ $= -14$	$v = -14$
3.6	$v = 15 + (-2.5) \times 4$ $= 15 - 10$ $= 5$	$v = 5$
substitution with a square		

3.7	$E = \frac{1}{2} \times 6 \times (-5)^2$ $= \frac{1}{2} \times 6 \times 25$ $= 75$	$E = 75$
3.8	$E = \frac{1}{2} \times 10 \times (-2)^2$ $= \frac{1}{2} \times 10 \times 4$ $= 20$	$E = 20$
3.9	$E = \frac{1}{2} \times 8 \times (-3)^2$ $= \frac{1}{2} \times 8 \times 9$ $= 36$	$E = 36$
3.10	$E = \frac{1}{2} \times 4 \times (-6)^2$ $= \frac{1}{2} \times 4 \times 36$ $= 72$	$E = 72$
3.11	$E = \frac{1}{2} \times 12 \times (-1)^2$ $= \frac{1}{2} \times 12 \times 1$ $= 6$	$E = 6$
3.12	$E = \frac{1}{2} \times 3 \times (-4)^2$ $= \frac{1}{2} \times 3 \times 16$ $= 24$	$E = 24$
simplifying expressions		
3.13		**(a)** $4x + 3y$ **(b)** $8a^4b^3$
3.14		**(a)** $6m + 4n$ **(b)** $18x^3y^3$
3.15		**(a)** $5c - 4d$ **(b)** $10p^3q^6$
3.16		**(a)** $10x + 4y$ **(b)** $18a^3b^4$
3.17		**(a)** $-3m + 7n$ **(b)** $14x^5y^3$
3.18		**(a)** $7p + 3q$ **(b)** $20a^4b^3$

Algebra 1 · Solving equations (Lesson 2.4)		
Q	**Working**	**Answer**
unknowns on both sides		
4.1	$4x - 5 = 11$ $4x = 16$ $x = 4$	$x = 4$
4.2	$4x + 4 = 24$ $4x = 20$ $x = 5$	$x = 5$
4.3	$3x - 3 = 12$ $3x = 15$ $x = 5$	$x = 5$
4.4	$4x + 2 = -10$ $4x = -12$ $x = -3$	$x = -3$
4.5	$6x - 9 = 15$ $6x = 24$ $x = 4$	$x = 4$
4.6	$3x + 8 = 23$ $3x = 15$ $x = 5$	$x = 5$
equations with fractions		
4.7	$2x + 5 = 21$ $2x = 16$ $x = 8$	$x = 8$
4.8	$4x - 3 = 25$ $4x = 28$ $x = 7$	$x = 7$
4.9	$5x + 2 = 12$ $5x = 10$ $x = 2$	$x = 2$

4.10	$3x + 4 = 10$ $3x = 6$ $x = 2$	$x = 2$
4.11	$2x - 1 = 15$ $2x = 16$ $x = 8$	$x = 8$
4.12	$6x + 3 = 15$ $6x = 12$ $x = 2$	$x = 2$
equations with brackets		
4.13	$8x - 4 = 3x + 21$ $5x = 25$ $x = 5$	$x = 5$
4.14	$5x - 10 = 4x + 6$ $1x = 16$ $x = 16$	$x = 16$
4.15	$9x + 6 = 4x + 36$ $5x = 30$ $x = 6$	$x = 6$
4.16	$8x - 6 = 3x + 9$ $5x = 15$ $x = 3$	$x = 3$
4.17	$6x - 6 = 4x + 8$ $2x = 14$ $x = 7$	$x = 7$
4.18	$8x + 4 = 6x + 2$ $2x = -2$ $x = -1$	$x = -1$